A Microphone Padding Approach for Speech Separation with Varying Number of Microphones

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Abstract. Many multi-microphone speech separation methods assume a microphone array with a fixed number of microphones arranged in a fixed geometry in training and testing. The trained models however often generalize poorly to unseen geometries and cannot be applied when the number of microphones differs in testing. To deal with this, transform-average-concatenate (TAC) and VarArray, which can train a single model that is invariant to microphone permutation and number, have been proposed. However, as they need to individually process each of the input microphones, the computation is prohibitively high, preventing their adoption in many real-world applications. In this context, we propose a simple yet effective microphone padding approach that unifies the input dimensionality by padding observed mixtures to a predefined maximum number of microphones so that the microphones can be processed all at once rather than individually, thereby reducing the computation. We propose some simple padding ways which not only achieve performance with TAC under the same computation but also achieve good generalization. Compared with TAC, experiments on ad-hoc dataset and fixed geometry array dataset have shown the validity of our methods.

Keywords: Speech separation · Microphone padding.

1 Introduction

Humans are capable of separating target speech of interest from a mixture with concurrent speech by multiple speakers, particularly in challenging environments such as meetings or social gatherings where multiple people could speak concurrently, a phenomenon commonly known as the cocktail party problem [3, 5, 17]. To enable this capability on machines, researchers employ microphone arrays to exploit spatial information for speech separation in multi-speaker environments,

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analogous to how humans use two ears to localize and segregate sound sources. This task is referred to as multi-channel speech separation, which has achieved significant progress by adopting deep learning [17]. Deep learning–based beamforming systems, commonly known as neural beamformers [8,13,25,27,34], have been widely applied to tackle this task. Two deep learning–based beamforming approaches have been extensively studied for multi-channel speech separation. One approach is to perform single-channel pre-separation followed by beamforming [1, 2, 4, 6, 7, 23, 28, 32]. This framework is invariant to the number and permutation of microphones, allowing it to accommodate varying microphone counts and arbitrary channel ordering. However, since it is not trained in an end-to-end manner, errors in the pre-separation stage may propagate to subsequent stages, and in addition, the pre-separation model does not leverage multi-channel information, potentially limiting overall performance. Another line of work directly estimates spatial filters in an end-to-end fashion using deep neural networks (DNN) [12, 15, 18, 20, 21, 24, 26, 29]. Although it offers a unified framework, it does not explicitly address the challenge of handling a variable number of microphones.

To mitigate the limitations of the cascaded "pre-separation + beamforming" pipelines and fully end-to-end filter prediction methods, FaSNet [10] proposed a hybrid solution. It directly estimates time-domain beamforming filters without explicitly assuming a fixed number or permutation of microphones. Specifically, FaSNet adopts a two-stage architecture, where a reference microphone is first enhanced through filter estimation, and subsequently, pairwise cross-channel features between the enhanced reference signal and each remaining microphone are used to estimate the corresponding beamforming filters. The estimated filters are then applied to their respective channels and summed to generate the final beamformed output. While this design enables permutation and microphone-number invariance, its two-stage structure still introduces dependency on the pre-separation quality, and the pairwise modeling strategy limits the ability to exploit global multi-channel information.

To address these limitations while preserving permutation and microphone-number invariance, the transform-average-concatenate (TAC) [11,30] framework was proposed as a simple yet effective multi-channel processing strategy. TAC enables permutation- and number-invariant modeling by first applying a shared transformation to the hidden representations of each microphone channel. The transformed representations are then averaged to obtain a global representation, which is further processed to enhance non-linear modeling capacity. Finally, the global representation is concatenated with each channel-wise feature and passed through another transformation to generate channel-dependent outputs. By combining global aggregation with channel-specific refinement, TAC allows the model to utilize information from all microphones. However, high computational cost remains a major concern. As each channel signal is processed independently during feature extraction, the overall computation increases drastically.

To reduce computation and at the same time achieve strong separation performance, we propose a microphone padding-based approach that unifies the input dimensionality across diverse microphone configurations. Given a pre-defined maximum number of microphones, we pad the number of input signals to this upper bound, thereby ensuring a fixed input dimensionality to the network. We propose four strategies for microphone padding, namely zero padding, zero random padding, repeat padding, and repeat random padding. Evaluation results demonstrate that the proposed approach achieves performance comparable to TAC when configured to use a similar or less amount of computation. Furthermore, the proposed microphone padding strategies exhibit better generalizability when trained on ad-hoc data but evaluated on data with fixed-geometry array configurations.

2 Microphone Padding

In a reverberant enclosure with C concurrent speakers recorded by a far-field microphone array with P microphones, the mixture at microphone p can be formulated, in the short-time Fourier transform (STFT) domain, as

$$X_{p,t,f} = \sum_{c=1}^{C} Y_{p,c,t,f} + N_{p,t,f}, \tag{1}$$

where t indexes T time frames, f indexes F frequency bins, and X_p, $Y_{p,c}$, and N_p respectively denote the STFT spectrograms of the mixture, reverberant speech of speaker c, and ambient noise captured at microphone p.

Without loss of generality, we designate microphone 1 as the reference microphone, and our goal is to estimate the reverberant speech $\{Y_{1,c}\}_{c=1}^{C}$ at the reference microphone 1 based on the multi-channel input mixture. We aim to train a single DNN-based speaker separation model that can handle a varying number of input microphones. Since many consumer electronics usually are not equipped with many microphones, we assume a maximum number of input microphones (denoted as M) that the model can handle. That is, P is always assumed to be smaller than M. Our idea is that for each input mixture, we can use microphone padding to pad the number of input microphones from P to M for both training and testing. This way, the input tensor to the DNN model always has the same size, which makes it much easier for us to train a single DNN model that can handle up to M input microphones.

In detail, the DNN is trained via complex spectral mapping [16,22–24], where the real and imaginary (RI) components of input mixture signals are stacked as input features to predict the RI components of target speech at the reference microphone. Following existing studies in speech separation, scale-invariant signal-to-distortion ratio (SI-SDR) [9] is employed as the loss function, and permutation invariant training [14, 31] is used to resolve the label-permutation problem. For an input mixture X with shape $P \times 2 \times T \times F$, where $P \leq M$ and the value of 2 means stacked RI components, we propose four microphone padding methods to

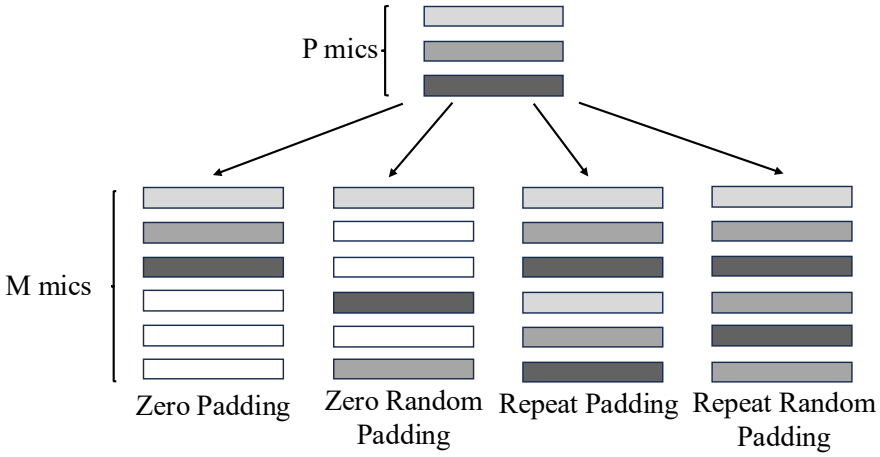


Fig. 1. Overview of microphone padding strategies.

pad X with shape $P \times 2 \times T \times F$ into X^{pad} with shape $M \times 2 \times T \times F$, ensuring a fixed input size for the DNN. The details of the four methods are described next. See Fig. 1 for an overview.

2.1 Zero Padding (ZP)

Motivated by the common practice of padding variable-length sequences to a fixed size in deep learning, we apply zero padding to standardize the number of input microphones to a predefined maximum M. This is akin to sequence padding in speech and natural language processing, where shorter inputs are extended with zeros to enable batch processing and fixed network dimensions.

Given an input with P available microphones with $P \leq M$, the signals from the first P channels are retained, while the remaining $M - P$ channels are filled with zeros. This operation converts inputs with a variable number of microphones into a fixed-channel representation compatible with the network. The padded input is expressed as

$$X_{t,f}^{\text{pad}} = \left[X_{1,t,f}, \ldots, X_{P,t,f}, 0, \ldots, 0\right] \in \mathbb{C}^{M}. \tag{2}$$

2.2 Zero Random Padding (ZRP)

While zero padding enforces a fixed input dimensionality, it preserves a deterministic microphone ordering, which may cause the model to associate specific spatial characteristics with the fixed microphone indices. Such a bias could degrade generalizability in ad-hoc scenarios where microphone positions vary.

To mitigate this issue, we propose zero random padding, where the reference microphone index (set to 1) is kept fixed, while the remaining $P-1$ microphones

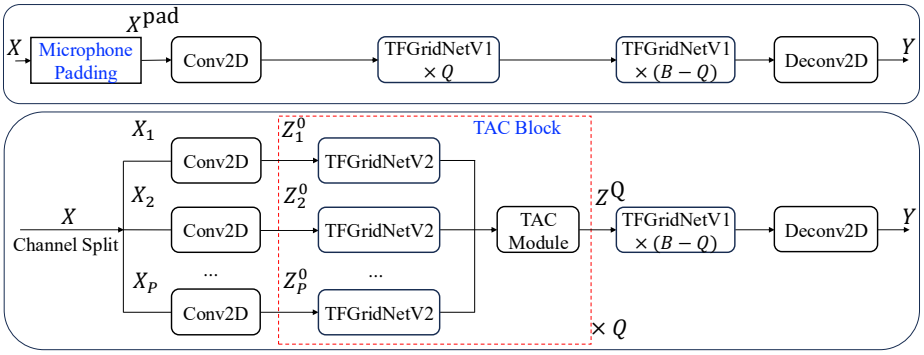


Fig. 2. Illustration of the proposed model (top) and TAC (bottom), which is configured to have a similar amount of computation to the proposed model.

are randomly assigned to the other $M-1$ slots without repetition. The unassigned $M-P$ slots are then filled with zeros. In detail, the padded input is constructed as

$$X_{t,f}^{\text{pad}} = \left[X_{1,t,f}, \ldots, 0, X_{i,t,f}, 0, \ldots\right] \in \mathbb{C}^M, \text{ for } i \in \{2, \ldots, P\}. \tag{3}$$

Compared with zero padding, which is deterministic, this randomization reduces the model's reliance on fixed channel indices and encourages learning position-invariant representations, which could be particularly beneficial for ad-hoc microphone configurations.

2.3 Repeat Padding (RP)

To avoid artificially silent channels introduced in zero padding and zero random padding, we further propose a repeat padding strategy. Given P input microphones, we repeatedly duplicate the available microphone signals until the number of channels reaches or exceeds M, and then retain only the first M channels. The repeat padding operation can be expressed as

$$X_{t,f}^{\text{pad}} = \left[X_{t,f}, X_{t,f}, \ldots\right]_{1:M} \in \mathbb{C}^M, \tag{4}$$

where $X_{t,f} = [X_{1,t,f}, \ldots, X_{P,t,f}] \in \mathbb{C}^P$. This strategy preserves the statistical properties and energy distribution of the original input without introducing zero-valued channels.

2.4 Repeat Random Padding (RRP)

In this strategy, the first P microphone signals are kept unchanged, while the remaining $M-P$ channels are randomly sampled from the existing P microphones with replacement. In other words, duplicated channels are allowed, and

each additional channel is independently drawn from the original P microphones. Formally, the padded input is constructed as

$$X_{t,f}^{\text{pad}} = \left[X_{t,f}, X_{\pi(1),t,f}, X_{\pi(2),t,f}, \ldots, X_{\pi(M-P),t,f}\right] \in \mathbb{C}^M, \tag{5}$$

where $\pi(i)$ is a random index sampled from the set $\{1, \ldots, P\}$. Compared with deterministic repeat padding, repeat random padding introduces stochasticity in channel duplication. This avoids zero-valued channels while encouraging the model to learn representations invariant to microphone ordering, thereby potentially improving robustness to varying microphone configurations.

3 Experimental Setup

3.1 Dataset

We validate the proposed algorithms based on a public dataset proposed in the TAC paper [11][3]. The dataset is designed for two-speaker separation in noisy-reverberant conditions, under ad-hoc and fixed-geometry array configurations. It is simulated based on the clean speech signals in LibriSpeech and noise signals in the *100 Nonspeech Sounds* corpus. Room impulse responses are synthesized using the image method with randomly sampled reverberation times (from 0.1 to 0.5 seconds) and room sizes. The energy level between the two speakers in each mixture is randomly sampled from the range $[-5, 5]$ dB. There are $20,000$, $5,000$ and $3,000$ mixtures for training, validation and testing, respectively. The average speaker overlap is 50%.

For the ad-hoc setting, the number of microphones varies from 2 to 6 with randomly sampled spatial layouts. The inter-microphone distances exhibit a wide range from as small as 0.05 m to as large as 11.0 m, with a mean of 3.0 m and a median of 2.7 m. Most microphone pairs are separated by several meters. Around 80% of the microphone pairs have distances between 1 m and 5 m, while 95% are within 6.2 m. Compact array configurations are rare in the dataset. These statistics indicate that the dataset is dominated by distributed microphone geometries rather than compact ones. The wide spatial coverage and substantial geometric variability provide diverse training conditions, which could improve the robustness of the trained models to different ad-hoc microphone layouts, particularly in distributed recording scenarios.

For the fixed-geometry setting, 6 microphones are arranged uniformly on a circular array with a fixed diameter of 10 cm. Different from the ad-hoc training set, where inter-microphone distances span a wide range, the test set is restricted to a compact-array configuration with distances in the range of $[5, 10]$ cm. The detailed simulation setup follows that in [11].

The reverberant speech signals at the reference microphone are used as the reference signals for metric computation. The evaluation metric is SI-SDR [9].

[3] The code for data generation is available at `https://github.com/yluo42/TAC/tree/master/data`.

3.2 System Configurations

We use TF-GridNet [19] as our backbone model. It models time–frequency (T-F) representations through joint temporal, spectral and spatial processing, and has demonstrated strong performance in speech separation tasks. The model is configured with the following hyperparameters. The embedding dimension D_1 is set to 128. The model is configured to have B TF-GridNet blocks. In each block, the kernel size I and stride size J for the Unfold and Deconv1D operations are both set to 1, the number of hidden units in each direction of the BLSTMs (denoted by H_1) is set to 200, the number of attention heads L is set to 4, and the number of output channels for the key and query projections in self-attention (denoted as E) is set to 256. The predefined maximum number M of input microphones is set to 6. We refer to this configuration of TF-GridNet block as **TF-GridNetV1**.

During training, the chunk size is set to 2 seconds. For STFT, the window size is set to 32 ms, hop size to 8 ms, and the square-root Hann window is used as the analysis window. All models are optimized using the SI-SDR loss [9] with an initial learning rate of 5×10^{-4}, which is reduced by half if the validation score does not improve for 4 consecutive epochs.

3.3 Baseline Systems

TAC [11] is considered as the baseline due to its popularity. See Fig. 2 for an illustration. To make a fair comparison between TAC and microphone padding, we configure them to both use TF-GridNet blocks, and in addition, as each TAC block needs to process each microphone individually, we use a lighter-weight TF-GridNet block (denoted as **TFGridNetV2**) so that each TAC block running TFGridNetV2 multiple times uses a similar amount of computation as each TFGridNetV1 block, which is run only once to process the microphones altogether. In each TFGridNetV2 block, the embedding dimension and hidden size are respectively set to $D_2 = 64$ and $H_2 = 100$, which are smaller than those in TFGridNetV1. All the other hyper-parameters are the same as TFGridNetV1.

In detail, given an input tensor $X \in \mathbb{R}^{P\times 2\times T\times F}$, each microphone slice $X_p \in \mathbb{R}^{2\times T\times F}$ is processed independently by using a TFGridNetV2 block. A 2D convolution (Conv2D) layer is first applied to each microphone to project the RI components into a D_2-dimensional feature space, producing a feature tensor $Z^0 = [Z_1^0, \ldots, Z_P^0] \in \mathbb{R}^{P\times D_2\times T\times F}$. The feature Z_p^0 at each microphone is then fed to a TFGridNetV2 block to capture local T-F dependencies. The resulting features are then passed to a TAC module to perform cross-channel spatial modeling, producing an output Z^1 with the same shape as Z^0. The TAC module is configured the same as that in [33], where the hidden dimension in the linear layer of the transform operation is set to $3 \cdot D_2$. This process is repeated Q times, where the output of the previous stage serves as the input to the next. After the TAC blocks, the embedding dimension is projected from D_2 to D_1 through a linear layer. Subsequently, a reference microphone (i.e., microphone 1) is selected from Z^Q, and used to select the reference feature for subsequent network modules, which consist of $B - Q$ TFGridNetV1 blocks.

Table 1. Results on ad-hoc arrays with various number of input microphones by using different strategies of microphone padding.

Padding method	P	Overlap ratio (%)				SI-SDRi (dB)
		≤ 25	$[25, 50)$	$[50, 75)$	≥ 75	
ZP	2	**24.3**	**20.8**	**19.3**	**16.9**	**20.3**
	4	**24.3**	**21.1**	**19.3**	**17.8**	**20.6**
	6	**24.3**	**21.5**	**18.8**	**17.4**	**20.5**
ZRP	2	23.2	19.9	18.0	15.7	19.2
	4	22.9	19.8	18.0	16.5	19.3
	6	23.3	20.0	17.8	16.3	19.3
RP	2	23.9	20.5	18.5	16.4	19.8
	4	23.7	20.3	18.6	17.0	19.9
	6	23.9	20.8	18.1	17.0	19.9
RRP	2	23.9	20.7	18.8	16.6	20.0
	4	23.9	20.7	19.0	17.3	20.2
	6	24.0	21.0	18.6	17.3	20.2

4 Evaluation Results

We validate our microphone padding strategies by training separation models on the ad-hoc training dataset and testing on the ad-hoc test dataset. In addition, we directly test the trained models on the fixed-geometry dataset to evaluate their generalizability to novel array geometry.

4.1 Results on Ad-hoc, Distributed Arrays

Table 1 reports the results of the four microphone padding strategies under the ad-hoc array setup (using $B = 4$ TFGridNetV1 blocks). Due to space limitations, we report results for three representative settings with 2, 4 and 6 input microphones. Although the proposed approach is simple and straightforward, it obtains strong performance, reaching 20.0 dB SI-SDRi. Among the padding strategies, zero padding achieves the highest SI-SDRi, zero random padding yields the lowest performance, and repeat padding and repeat random padding obtain similar performance in between. The performance difference between the deterministic and random variants suggests that introducing excessive randomness during padding does not necessarily improve robustness on the matched ad-hoc test set. In particular, the stronger performance of zero padding indicates that explicitly distinguishing the observed micßrophones from the padded positions is sufficient for the network to handle a variable number of input channels. Meanwhile, the comparable results of RP and RRP show that duplicating available microphone signals does not substantially degrade separation performance, even though it introduces redundant observations.

Table 2. Performance comparison of microphone padding and TAC on ad-hoc arrays with various number of input microphones using similar GMACs.

Method	GMACs	P	Overlap ratio (%)				SI-SDRi (dB)
			≤ 25	$[25, 50)$	$[50, 75)$	≥ 75	
TAC $1+4$	362.0	2	21.1	17.5	15.9	13.8	17.1
	407.1	4	21.1	17.7	16.1	14.4	17.3
	452.3	6	21.0	18.2	15.8	14.0	17.2
TAC $2+3$	331.0	2	23.4	19.5	18.1	16.1	19.2
	421.1	4	24.1	21.1	19.2	17.7	20.5
	511.2	6	24.5	21.7	19.0	17.7	20.7
TAC $2+4$	409.4	2	23.9	20.1	18.6	16.7	19.8
	499.5	4	24.4	**21.6**	**19.6**	**18.3**	**21.0**
	589.6	6	**25.0**	**22.2**	**19.3**	**18.3**	**21.2**
RRP-4	314.8	2	23.9	20.7	18.8	16.6	20.0
		4	23.9	20.7	19.0	17.3	20.2
		6	24.0	21.0	18.6	17.3	20.2
RRP-5	393.2	2	**24.7**	**21.6**	**19.7**	**17.5**	**20.9**
		4	**24.5**	**21.6**	19.4	18.0	20.9
		6	24.6	21.9	19.2	17.8	20.9

Table 2 reports the results of TAC under the ad-hoc array setup. Since the computation of the same TAC model increases when the number of input microphones increases, due to space limitations, we only report its GMACs under conditions of using 2, 4 and 6 microphones as input. Here, "TAC $1+4$" means $B = 1+4, Q = 1$ and "RRP-4" means $B = 4$. Other models use the same naming rule. We observe that the performance of TAC increases with the number of input microphones, especially for the speaker overlap in the range of $[0, 50\%)$. In comparison, RRP has similar performance when the number of input microphones increases. In addition, RRP obtains comparable performance to TAC, with similar or less computation. These results also reveal different sensitivities of the two approaches to the number of available microphones. TAC benefits more clearly from additional microphones because its channel communication modules explicitly aggregate information across the observed channels. In contrast, the relatively stable performance of RRP indicates that the padding-based model is less dependent on increasing the number of spatial observations. The advantage of TAC is more evident in low-overlap conditions, whereas its improvement becomes smaller as the overlap ratio increases, suggesting that severe speaker overlap remains a major source of difficulty even when more microphone observations are available.

Table 3. Results of microphone padding on 6-mic fixed-geometry array.

Padding method	Overlap ratio (%)				SI-SDRi (dB)
	≤ 25	$[25, 50)$	$[50, 75)$	≥ 75	
ZP	**20.7**	18.0	16.0	**14.7**	17.3
ZRP	19.4	16.6	14.4	13.1	15.9
RP	**20.7**	18.0	16.0	14.6	17.3
RRP	20.6	**18.1**	**16.1**	**14.7**	**17.4**

Table 4. Performance comparison of microphone padding and TAC on 6-mic fixed-geometry array using similar GMACs.

Method	GMACs	Overlap ratio (%)				SI-SDRi (dB)
		≤ 25	$[25, 50)$	$[50, 75)$	≥ 75	
TAC $1 + 4$	452.3	19.3	16.2	14.3	12.8	15.6
TAC $2 + 3$	511.2	20.8	17.9	15.9	14.4	17.2
TAC $2 + 4$	589.6	19.7	17.4	15.4	14.2	16.7
RRP-4	314.8	20.6	18.1	16.1	14.7	17.4
RRP-5	393.2	**21.2**	**18.7**	**16.7**	**15.3**	**18.0**

4.2 Results on Unseen Compact Array

Table 3 reports the separation results on the fixed-geometry array dataset, where the separation models are trained on the ad-hoc dataset. Each model has $B = 4$ TFGridNetV1 blocks. We observe that all the microphone padding strategies achieve strong performance, and RRP shows the best generalization ability, even though it does not obtain the highest performance on the ad-hoc dataset. Notably, the advantage of RRP is consistent across different overlap-ratio ranges, although its overall improvement over ZP and RP is relatively small. This result suggests that the random repetition used during training may reduce the model's dependence on a particular correspondence between input positions and physical microphones. Consequently, RRP can be interpreted as a form of channel-level augmentation, which exposes the model to more diverse padded input patterns and improves its robustness when the microphone geometry changes from a distributed ad-hoc array to a compact fixed array. Nevertheless, all methods experience a gradual performance degradation as the overlap ratio increases, confirming that highly overlapped speech remains challenging under unseen array geometries.

Table 4 compares the results of TAC and the RRP strategy under similar or less computation on the 6-microphone fixed-geometry array dataset. Although "TAC $2+4$" obtains strong performance when tested on the ad-hoc test dataset, "RRP-5" achieves better performance using less computation, indicating that the RRP strategy has strong generalizability.

5 Conclusions

We have proposed a microphone padding approach for multi-channel speech separation with varying number of microphones. It assumes a maximum number of microphones the model can handle, and proposes various microphone padding strategies to ensure that the number of input signals to the model is always the same. Although extremely simple, it obtains strong performance on a speech separation dataset with ad-hoc arrays, and exhibits reasonable generalizability with unseen compact arrays.

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